

***Abstract.** A game of squash can be played either to nine or to fifteen points. If you think you are a better player than your opponent, then which game—the shorter version, or lengthier version—provides you a higher probability of winning? Suppose, instead, that you are the weaker player. Which game is your best bet now? If you believe that you would favor the same game length in either case, then consider this theorem from probability theory: the larger the sample of rounds (i.e., fifteen rounds versus nine rounds), the greater likelihood of achieving the expected outcome (i.e., victory to the stronger player). So, if you believe you are the stronger player, then you should prefer the longer game; believing yourself to be the weaker player should produce a preference for the shorter game. Intuitively, though, victory over an opponent in either a nine-point or fifteen-point match would strike many people as equally representative of one's aptitude at squash. This is an example of sample-size neglect bias.¹*

The concept of permitting the game “to go longer” in order to increase the probability that the stronger player wins can also apply to investing, where it is called time diversification, which refers to the idea that investors should spread their assets across ventures operating according to a variety of market cycles, giving their allocations plenty of time to work properly. Time diversification helps reduce the risk that an investor will be caught entering or abandoning a particular investment or category at a disadvantageous point in the economic cycle. It is particularly relevant with regard to highly volatile investments, such as stocks and long-term bonds, whose prices can fluctuate in the short term. Holding onto these assets for longer periods of time can soften the effects of such fluctuations. Conversely, if an investor cannot remain in a volatile investment over a relatively long time period, he or she should avoid the investment. Time diversification is less important when considering relatively stable investments, such as certificates of deposit, money market funds, and short-term bonds.

Time diversification also comes into play when investing or withdrawing large sums of money from a specified niche within an allocation. In general, it is best to move these amounts gradually over time, rather than all at once, to reduce risk. Borrowed from Kenneth Fisher and Meir Statman,² Figures 7.1 and 7.2 show a pair of graphic models illustrating expected average annual returns over a 1-year and a 30-year horizon, respectively.